

Web Sistemak

1. Praktika: IoT bezeroa

Kudeaketaren eta Informazio Sistemen
Informatika Gradua

Sistemen Ingeniaritza eta Automatika Saila

8. Pausua: ThinkSpeak

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Universidad
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Gaiaren egitura

- ThingSpeak IoT plataformaren aurkezpena.
- Datuak kanaletan biltegitratzea
- Kanal baten kudeaketa
- ThingSpeak-en REST API-a
- Datuak bidaltzeko adibideak:
 - Datuak inprimaki formatuan bidali eta erantzun huts bat jaso.
 - Datuak inprimaki formatuan bidali eta erantzun bat jaso JSON formatuan.
 - Datuak JSON formatuan bidali eta erantzun bat jaso XML formatuan.

ThingSpeak IoT plataforma

- ThingSpeak zerbitzuak datuak biltegitatzeko zerbitzua eta interfazeak (HTTP eta MQTT) eskaintzen ditu.
- Doan erabiltzeko lizentzia urtero berritzen da automatikoki, eta 3.000.000 mezu bidaltzeko aukera ematen du.

My Account

License Type: Free

Free Messages	Messages Remaining	Expiration Date
3.000.000	2.940.461	04 Dec 2019

To purchase a license:

[How to Buy](#)

Recommended maximum daily usage based on your annual capacity: **8.219 messages**

For questions about exceeding suggested daily usage rate or for purchasing new units, see the [Licensing FAQ](#)



Datuak kanaletan biltegitratzea

Datuak erabiltzaile-kontuei lotutako "kanaletan" gordetzen dira. Kanalak publikoak edo pribatuak izan daitezke.

My Channels

New Channel Search by tag

Name	Created	Updated
 Weather Station in Amurrio, Basque Country temperature, relative humidity, atmospheric pressure, DS18B20, HH10D, BMP180, Arduino, xbee Private Public Settings Sharing API Keys Data Import / Export	2016-11-28	2018-09-03 18:59
 Mobile weather station PiE Private Public Settings Sharing API Keys Data Import / Export	2018-07-24	2018-12-14 14:42
 GSM MQTT client Private Public Settings Sharing API Keys Data Import / Export	2018-07-26	2018-07-30 12:02
 Prueba Private Public Settings Sharing API Keys Data Import / Export	2019-01-17	2019-01-17 20:44

Kanal publikoak

Kanal pribatuak

Datuak kanaletan biltegitratzea

- Kanal batean 8 datu-eremu gorde daitezke.
- Eremuei esleitzen zaizkien etiketak deskribatzaileak dira. Eremuen izenak honako hauek dira: field1, field2, field3...

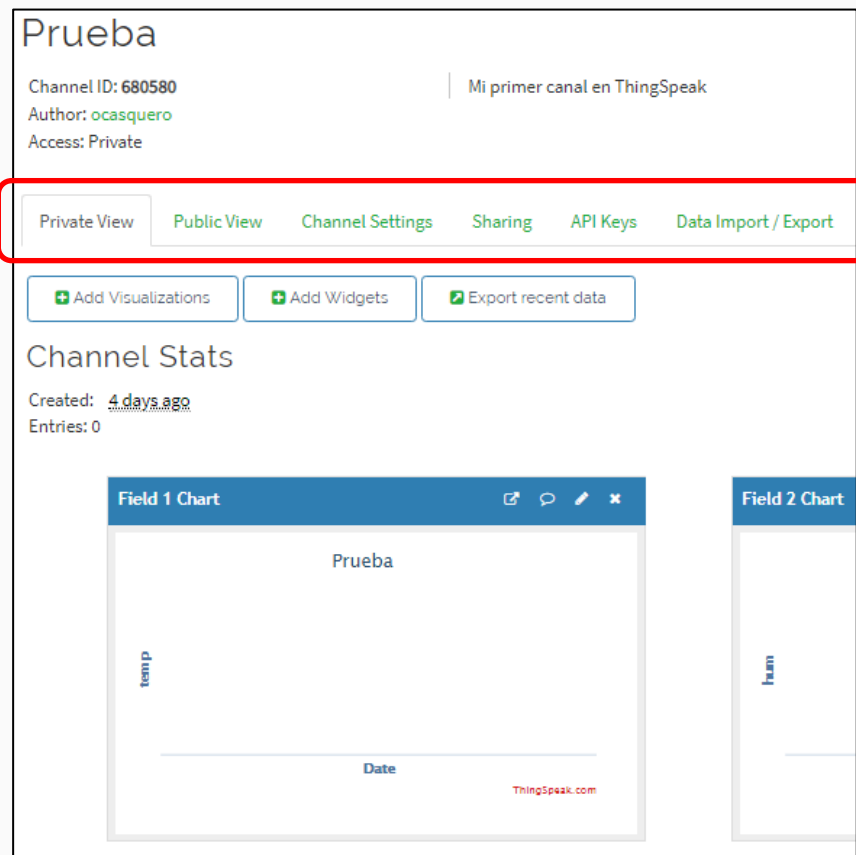
New Channel

Name

Description ThingSpeak"/>

Field 1	temp	☒
Field 2	hum	☒
Field 3		☐
Field 4		☐
Field 5		☐
Field 6		☐
Field 7		☐
Field 8		☐

Kanal baten kudeaketa



- *Private view y Public view* erlaitzek datuak bistartzeko aukera ematen dute.
- *Channel settings* erlaitzak ondoko ekintzak ahalbidetzen ditu:
 - Datu-eremuak editatzea
 - Kanala hustea
 - Kanal ezabatzea
- *Sharing* kanala partekatzeko
- *API keys*:
 - *Write API key* kanalean datuak idazteko.
 - *Read API key* kanal pribatu batetik datuak irakurtzeko.
- *Data import/export*

ThingSpeak-en REST API-a

- ThingSpeak-en REST API-a ThingSpeak plataformak HTTP protokoloaren bidez kanalak kudeatzeko eskaintzen dituen eragiketa/zerbitzuen multzoak osatzen du.
- ThingSpeak plataformaren REST API-aren dokumentazioa: <https://es.mathworks.com/help/thingspeak/rest-api.html>

The screenshot shows the 'REST API' section of the ThingSpeak documentation. It includes an introduction to REST API calls and a 'RESTful API Reference' table. Annotations highlight specific parts of the interface:

- Zerbitzu motak** (Service types) points to the 'Read Data', 'Write Data', 'View Channels and Channel Settings', and 'Create and Delete' categories.
- Irakurketa eragiketak** (Reading operations) points to the 'Read Data', 'Read Field', 'Read Status', 'Read Last Entry', 'Read Last Field Entry', and 'Read Last Status' rows in the table.

RESTful API Reference	
▼ Read Data	
Read Data	Read data from all fields in a channel with HTTP GET
Read Field	Read data from a single field of a channel with HTTP GET
Read Status	Read status field of a channel with HTTP GET
Read Last Entry	Read the last entry in a channel with HTTP GET
Read Last Field Entry	Read the last entry in a field of a channel with HTTP GET
Read Last Status	Read the last status of a channel with HTTP GET
> Write Data	
> View Channels and Channel Settings	
> Create and Delete	

ThingSpeak-en REST API-a

Dokumentazioak datuak bidaltzeko HTTP eskaera nola eraiki behar den adierazten du.

Write Data
Update channel data with HTTP GET or POST

Request

HTTP Method
POST or GET

URL
`https://api.thingspeak.com/update.<format>`

URL Parameters

Name	Description
<format>	(Required) Format for the HTTP response, specified as blank, json, or xml.

Example: `https://api.thingspeak.com/update.json`

Body

Name	Description	Value-Type
api_key	(Required) Specify Write API Key for this specific channel. The Write API Key can optionally be sent via a THINGSPEAKAPIKEY HTTP header. The Write API Key is found on the API Keys tab of the channel view.	{Write - API - Key string}
field<X>	(Optional) Field X data, where X is the field ID.	<any>
lat	(Optional) Latitude in degrees.	<decimal>
long	(Optional) Longitude in degrees.	<decimal>
elevation	(Optional) Elevation in meters.	<integer>
status	(Optional) Status update message.	<string>
twitter	(Optional) Twitter® username linked to ThingTweet.	<string>
tweet	(Optional) Twitter status update.	<string>
created_at	(Optional) Date when feed entry was created, in ISO 8601 format, for example: 2014-12-31 23:59:59. Must be unique within channel. Time zones can be specified via the timezone parameter.	<datetime>

Content-Type
Content-Type is required only for the POST method, not for the GET method.
application/x-www-form-urlencoded for most updates.
application/json for updates in JSON format.

ESKAERA ADIBIDEAK:

Datuak inprimaki formatuan bidali eta erantzun huts bat jaso (OK bada, erregistroaren ID-a itzultzen du; errorrik bada, "0" itzultzen du)

POST /update HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/x-www-form-urlencoded
Content-Length: 44

api_key=OQ1CVZKQX7HV9J8Y&field1=22&field2=47

Datuak inprimaki formatuan bidali eta erantzuna JSON formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorrik bada, "0" itzultzen du)

POST /update.json HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/x-www-form-urlencoded
Content-Length: 44

api_key=OQ1CVZKQX7HV9J8Y&field1=22&field2=47

Datuak JSON formatuan bidali eta erantzuna XML formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorrik bada, "0" itzultzen du)

POST /update.xml HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/json
Content-Length: 58

{"api_key":"OQ1CVZKQX7HV9J8Y","field1":"22","field2":"47"}



Datuak bidaltzea kanal batean biltegitratzeko

Adibidea 1: Datuak inprimaki formatuan bidali eta erantzun huts bat jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

ESKAERA ADIBIDEAK:

Datuak inprimaki formatuan bidali eta erantzun huts bat jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

```
POST /update HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/x-www-form-urlencoded
Content-Length: 44

api_key=OQ1CVZKQX7HV9J8Y&field1=22&field2=47
```

Datuak inprimaki formatuan bidali eta erantzuna JSON formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

```
POST /update.json HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/x-www-form-urlencoded
Content-Length: 44

api_key=OQ1CVZKQX7HV9J8Y&field1=22&field2=47
```

Datuak JSON formatuan bidali eta erantzuna XML formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

```
POST /update.xml HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/json
Content-Length: 58
```

```
{"api_key": "OQ1CVZKQX7HV9J8Y", "field1": "22", "field2": "47"}
```

Response

```
Raw Headers Hex
HTTP/1.1 200 OK
Content-Type: text/html; charset=utf-8
Content-Length: 1
Connection: close
Status: 200 OK
X-Frame-Options: SAMEORIGIN
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, OPTIONS, DELETE, PATCH
Access-Control-Allow-Headers: origin, content-type, X-Requested-With
Access-Control-Max-Age: 1800
ETag: "1679091c5a880faf6fb5e6087eb1b2dc"
Cache-Control: max-age=0, private, must-revalidate
Set-Cookie: request_method=POST; path=/
X-Request-Id: 17abd716-ca4f-4c30-920c-46ac4e3ba3f3
X-Runtime: 0.090177
X-Powered-By: Phusion Passenger 4.0.57
Date: Tue, 22 Jan 2019 11:05:22 GMT
Server: nginx/1.9.3 + Phusion Passenger 4.0.57
```

Datuak bidaltzea kanal batean biltegitratzeko

Adibidea 2: Datuak inprimaki formatuan bidali eta erantzuna JSON formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

ESKAERA ADIBIDEAK:

Datuak inprimaki formatuan bidali eta erantzun huts bat jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

```
POST /update HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/x-www-form-urlencoded
Content-Length: 44
```

```
api_key=OQ1CVZKQX7HV9J8Y&field1=22&field2=47
```

Datuak inprimaki formatuan bidali eta erantzuna JSON formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

```
POST /update.json HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/x-www-form-urlencoded
Content-Length: 44
```

```
api_key=OQ1CVZKQX7HV9J8Y&field1=22&field2=47
```

Datuak JSON formatuan bidali eta erantzuna XML formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

```
POST /update.xml HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/json
Content-Length: 58
```

```
{"api_key": "OQ1CVZKQX7HV9J8Y", "field1": "22", "field2": "47"}
```

Response

Raw

Headers

Hex

```

HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Connection: close
Status: 200 OK
X-Frame-Options: SAMEORIGIN
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, OPTIONS, DELETE, PATCH
Access-Control-Allow-Headers: origin, content-type, X-Requested-With
Access-Control-Max-Age: 1800
ETag: "c2841a2a7a7457a8b37f50a7b112e2d1"
Cache-Control: max-age=0, private, must-revalidate
Set-Cookie: request_method=POST; path=/
X-Request-Id: 0723bd68-73f8-493e-a09e-32f50a06c73f
X-Runtime: 0.033213
X-Powered-By: Phusion Passenger 4.0.57
Date: Tue, 22 Jan 2019 11:10:33 GMT
Server: nginx/1.9.3 + Phusion Passenger 4.0.57
Content-Length: 246

{"channel_id":680580,"created_at":"2019-01-22T11:10:33Z","entry_id":7,"field1":"22","field2":"47","field3":null,"field4":null,"field5":null,"field6":null,"field7":null,"field8":null,"latitude":null,"longitude":null,"elevation":null,"status":null}
```



Datuak bidaltzea kanal batean biltegitratzeko

Adibidea 3: Datuak JSON formatuan bidali eta erantzuna XML formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

ESKAERA ADIBIDEAK:

Datuak inprimaki formatuan bidali eta erantzun huts bat jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

POST /update HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/x-www-form-urlencoded
Content-Length: 44

api_key=OQ1CVZKQX7HV9J8Y&field1=22&field2=47

Datuak inprimaki formatuan bidali eta erantzuna JSON formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

POST /update.json HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/x-www-form-urlencoded
Content-Length: 44

api_key=OQ1CVZKQX7HV9J8Y&field1=22&field2=47

Datuak JSON formatuan bidali eta erantzuna XML formatuan jaso (OK bada, erregistroaren ID-a itzultzen du; errorik bada, "0" itzultzen du)

POST /update.xml HTTP/1.1
Host: api.thingspeak.com
Content-Type: application/json
Content-Length: 58

{"api_key": "OQ1CVZKQX7HV9J8Y", "field1": "22", "field2": "47"}

Response

Raw Headers Hex XML

```
HTTP/1.1 200 OK
Content-Type: application/xml; charset=utf-8
Connection: close
Status: 200 OK
X-Frame-Options: SAMEORIGIN
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, OPTIONS, DELETE, PATCH
Access-Control-Allow-Headers: origin, content-type, X-Requested-With
Access-Control-Max-Age: 1800
ETag: "331bb5526b094f19368308ddb73af7ce"
Cache-Control: max-age=0, private, must-revalidate
Set-Cookie: request_method=POST; path=/
X-Request-Id: cc7c47e8-8d8c-450e-8d8e-a0cbe9b1afc8
X-RunTime: 0.042861
X-Powered-By: Phusion Passenger 4.0.57
Date: Tue, 22 Jan 2019 11:12:39 GMT
Server: nginx/1.9.3 + Phusion Passenger 4.0.57
Content-Length: 489

<?xml version="1.0" encoding="UTF-8"?>
<feed>
  <channel-id type="integer">680580</channel-id>
  <created-at type="dateTime">2019-01-22T11:12:39Z</created-at>
  <entry-id type="integer">8</entry-id>
  <field1>22</field1>
  <field2>47</field2>
  <field3 nil="true"/>
  <field4 nil="true"/>
  <field5 nil="true"/>
  <field6 nil="true"/>
  <field7 nil="true"/>
  <field8 nil="true"/>
  <latitude nil="true"/>
  <longitude nil="true"/>
  <elevation nil="true"/>
  <status nil="true"/>
</feed>
```

